Post-Quantum Lattice-Based Cryptography: Empirical Evaluation of CRYSTALS-Kyber a

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Abstract

With NIST finalizing post-quantum cryptography standards, transitioning enterprise networks to lattice-based primitives is paramount. This paper presents an empirical performance benchmark of CRYSTALS-Kyber key encapsulation and CRYSTALS-Dilithium signatures across heterogeneous server, cloud, and embedded ARM architectures. While Kyber-768 requires 1.8x larger memory buffers during key generation compared to RSA-2048, it achieves 12.4x faster encapsulation speed and imperviousness to Shor algorithm attacks. However, public key and ciphertext transmission induces TCP packet fragmentation on legacy networking middleboxes. We evaluate mitigation strategies and outline deployment guidelines for zero-trust environments.

1. Objective and Research Scope

The objective of this research is to measure the actual network throughput, memory overhead, and side-channel resistance of NIST FIPS 203 (ML-KEM / Kyber) and FIPS 204 (ML-DSA / Dilithium). As organizations respond to quantum decryption threats, practical adoption hurdles must be separated from theoretical security proofs. We investigate whether lattice-based schemes can seamlessly replace RSA-2048 and ECDSA without degrading TLS 1.3 handshake latency under real-world congestion.

2. Experimental Methodology

We executed 500,000 TLS 1.3 handshakes across three target testbeds: Intel Xeon Platinum 8380 server clusters, Raspberry Pi 4 (ARM and STM32 Cortex-M4 microcontrollers with 192 KB RAM. We recorded memory allocation watermarks, CPU clock cycles using RDTSC, packet fragmentation rates across simulated WAN links with 40ms round-trip latency and 0.5% packet drop rates.
Constant-time polynomial multiplication was verified using valgrind-based side-channel leakage detection.

3. Key Findings and Benchmark Comparison

The experimental benchmark yielded the following critical observations:

1. Computational Speed: Kyber-768 key encapsulation completed in 38.2 microseconds on Intel Xeon, compared to 475 microseconds fo
   yielding an impressive 12.4x speed acceleration.
2. Key and Ciphertext Size: Kyber-768 public keys are 1,184 bytes (versus 256 bytes for RSA-2048 and 64 bytes for ECDSA P-256).
3. Transmission Penalty: In WAN environments, Dilithium-3 signatures (3,293 bytes) triggered IP fragmentation in 14.8% of packets acros
   firewalls, causing a 28% increase in tail P99 latency.
4. Agreement with Prior Studies: In strong agreement with quantum threat models, our security verification confirms complete mathematic
   immunity against Shor algorithm attacks based on the Module Learning With Errors (M-LWE) hardness problem.

4. Limitations and Embedded Device Constraints

We observed several prominent practical limitations:

- On low-power Cortex-M4 IoT microcontrollers, the 1.8x memory footprint expansion caused stack overflows without dedicated RTOS me
- Network middlebox packet drop rates remain an unresolved obstacle in enterprise legacy architectures without MTU renegotiation.
- The study did not test resistance against physical electromagnetic fault-injection side channels on unshielded microcontrollers.

5. Conclusions and Future Research Scope

We conclude that lattice-based cryptography is completely viable and superior for cloud and enterprise services today.

However, we disagree with proposals advocating mandatory instant adoption across all low-power IoT devices prior to hardware accelera

Future research must explore hybrid post-quantum schemes combining classical and lattice algorithms, alongside zero-fragmentation pac